UMAT-OTI workbench

Generate and verify derivatives of an Abaqus UMAT, without editing the subroutine.

1. Source | 2. Material model | **3. Derivatives and loading** | 4. Run and results

m3_j2 · fixed form · 1 routine · 0 local Newton solves · NTENS 6 · NSTATV 1 · 4 constants

3. Derivatives and loading

Choose what to compute, and how to load it.

What should be computed?

- [ ] Consistent tangent · `DDSDDE` ?
- [x] Internal Jacobian of the local solve · `FJAC / DF` ?
- [ ] Higher-order stress derivatives · `orders 2 and above` ?
- [x] Stress sensitivity to material parameters · `DSIGMA_DP` ?
- [x] State sensitivity to material parameters · `DSTATEV_DP` ?

How should the material be loaded?

Loading history ?

Uniaxial strain, 20 steps of 1e-4 (small strain)

Stretches along one axis in 20 equal steps. For a steel-like model this crosses the yield point part way through, so both the elastic and the inelastic response are exercised.

20 increments · provenance: parameter_sensitivity/loading_paths.json default

Before you run

UNSUPPORTED **Internal Jacobian of the local solve** — This source integrates its law without a local Newton iteration, so it has no internal Jacobian to extract.

The rest of the request can still run: Stress sensitivity to material parameters, State sensitivity to material parameters

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